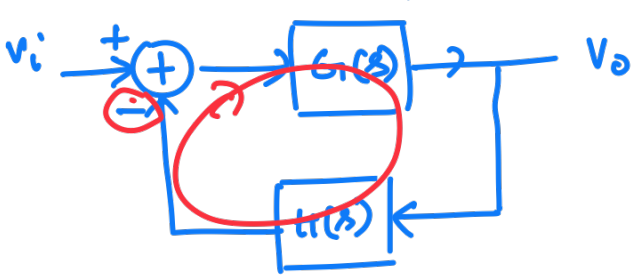
Lecture #14



$$\frac{V_o(s)}{V_i(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

$1 + LG = 1 - 1 = 0$

$$= \frac{1}{H(s)} \cdot \frac{G(s)H(s)}{1 + G(s)H(s)}$$

$1 + G(s)H(s)$: $D(s)$

$$G(s) = \frac{N_g(s)}{D_g(s)}, \quad H(s) = \frac{N_h(s)}{D_h(s)}$$

$$1 + G(s)H(s) = 1 + \frac{N_g(s)N_h(s)}{D_g(s)D_h(s)} = \frac{D_g(s)D_h(s) + N_g(s)N_h(s)}{D_g(s)D_h(s)}$$

$D(s)$ second order tf.
$\geq$ third order

if $D(s)$ has right half plane roots, system is unstable.

"Nyquist Stability Criterion"

⇓ Simplified

Loop gain has only poles. → [loop gain and phase margin concept.]

$D(s) = 1 + LG$

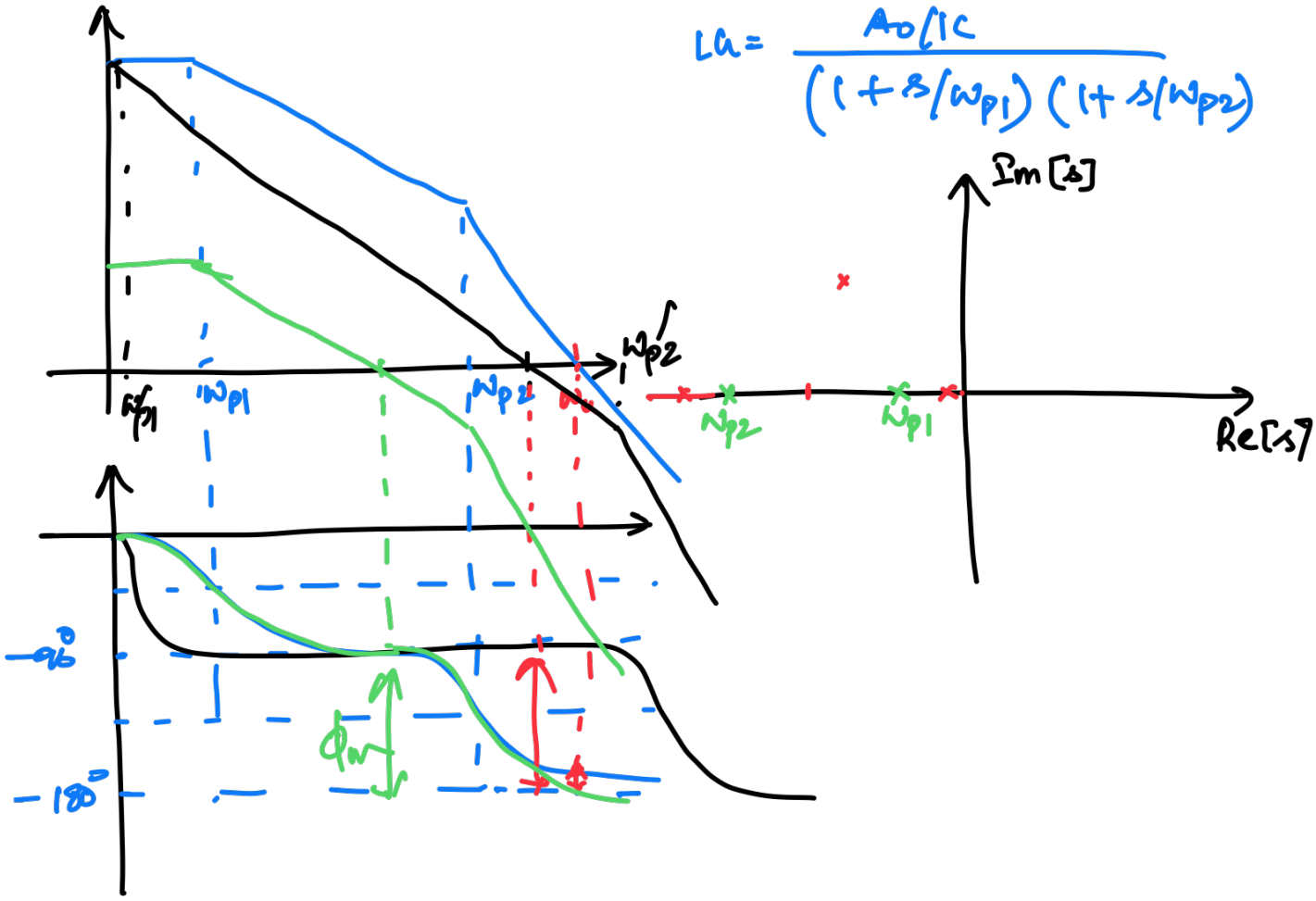
$\angle LG(j\omega)| = 180^\circ$ at $\omega = \omega_0$

$|LG(j\omega_0)| \geq 1$

$1 - |LG|$ at $\omega = \omega_0$

$\frac{1 + LG}{=0}$ ⇒ $|LG| > 1$ and $\angle LG = 180^\circ$

$$LG = \frac{A_0 / IC}{(1 + s/\omega_{p1})(1 + s/\omega_{p2})}$$





$$LG = \frac{A_{02}}{K(1 + s/\omega_{p1})(1 + s/\omega_{p2})}$$

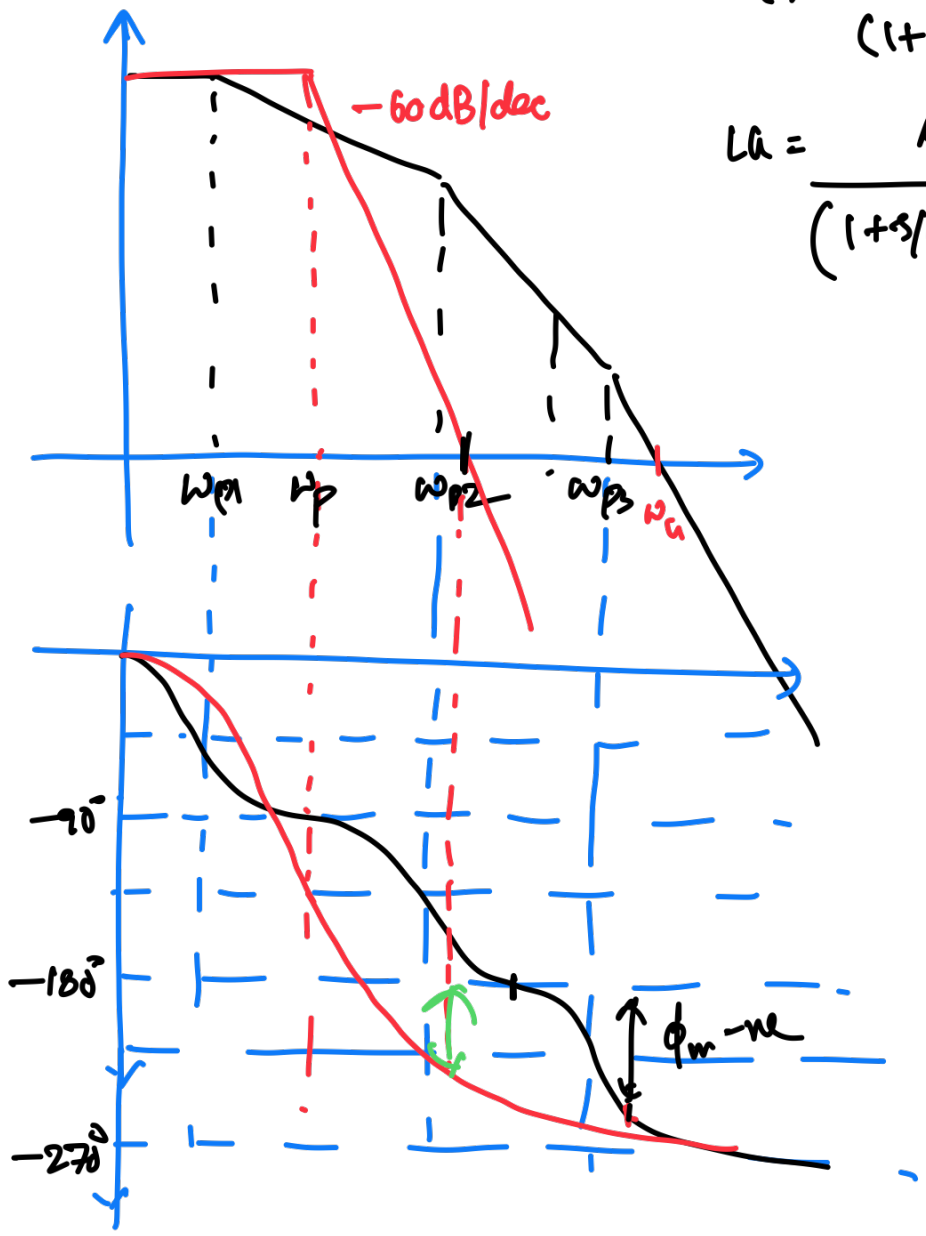
$$Q_p = \frac{\sqrt{\frac{A_0}{K} + 1}}{\sqrt{\frac{\omega_{p1}}{\omega_{p2}}} + \sqrt{\frac{\omega_{p2}}{\omega_{p1}}}}$$

} closed loop tf.

- High quality factor (Q_p)
  Low damping coefficient (ζ) } → low phase margin

$$A(s) = \frac{A_{03}}{(1+s/\omega_{p1})(1+s/\omega_{p2})(1+s/\omega_{p3})}$$

$$L(s) = \frac{A_{03}/K}{(1+s/\omega_{p1})(1+s/\omega_{p2})(1+s/\omega_{p3})}$$



$$A(s) = \frac{A_{03}}{(1+s/\omega_{p3})^3}$$

$$L(s) = \frac{A_{03}/K}{(1+s/\omega_{p3})^3} \approx \frac{A_{03}/K}{(\omega_u/\omega_{p3})^3}$$

$$\omega_u \approx \omega_{p3}\left(\frac{A_{03}}{K}\right)^{1/3}$$

$$D(s) = 1 + L(s) = 1 + \frac{A_{03}/K}{(1+s/\omega_{p3})^3} = 0$$

$$(1+s/\omega_{p3})^3 + \frac{A_{03}}{K} = 0$$

$$\left(1 + s/\omega_{p3}\right)^3 = (-1)\frac{A_{03}}{K}$$

$$\left(1 + s/\omega_{p3}\right) = \left[e^{j(2p+1)\pi}\right]^{1/3}\left(\frac{A_{03}}{K}\right)^{1/3}$$

$$s = +\omega_{p3}\left[-1 + e^{j(2p+1)\pi/3}\left(\frac{A_{03}}{K}\right)^{1/3}\right]$$

$$\downarrow$$

$e^{j\pi/3}$ $e^{j3\pi/3}$ $e^{j5\pi/3}$

$$\left(\frac{1}{2} + j\frac{\sqrt{3}}{2}\right) \quad -1 \quad \left(\frac{1}{2} - j\frac{\sqrt{3}}{2}\right)$$

$$s_1 = \omega_{p3}\left[-1 + (-1)\left(\frac{A_{03}}{K}\right)^{1/3}\right]$$

$$s_2 = \omega_{p3}\left[-1 + \left(\frac{1}{2} + j\frac{\sqrt{3}}{2}\right)\left(\frac{A_{03}}{K}\right)^{1/3}\right]$$

$$s_3 = \omega_{p3}\left[-1 + \left(\frac{1}{2} - j\frac{\sqrt{3}}{2}\right)\left(\frac{A_{03}}{K}\right)^{1/3}\right]$$

$$-1 + \frac{1}{2}\left(\frac{A_{03}}{K}\right)^{1/3} > 0 \quad (\text{R.H.P})$$

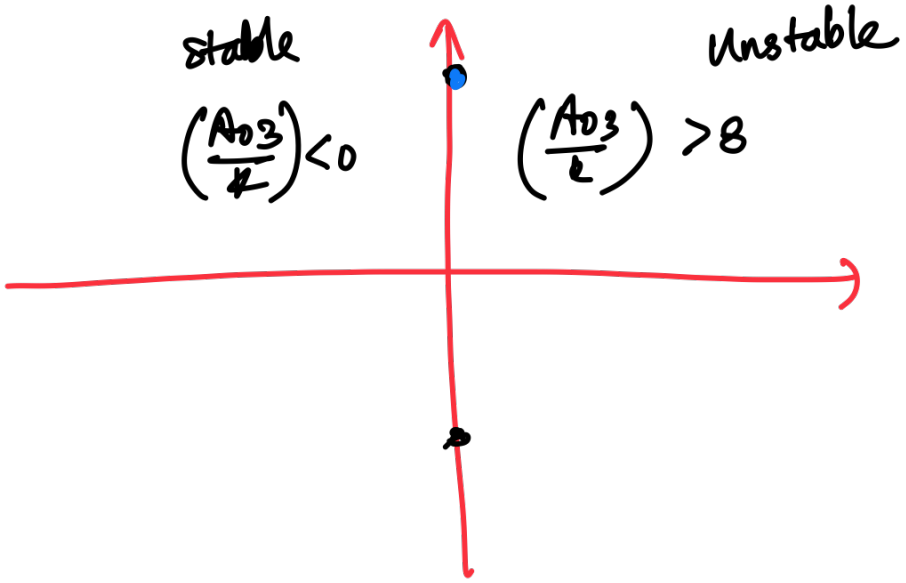
$$-1 + \frac{1}{2}\left(\frac{A_{03}}{K}\right)^{1/3} = 0$$

$$\frac{A_{03}}{K} = 8$$

For L.H.P, $\quad -1 + \frac{1}{2}\left(\frac{A_{03}}{K}\right)^{1/3} < 0$

$$\frac{A_{03}}{K} < 8$$

$$(s-s_1)(s-s_2)(s-s_3)$$



$$s_1 = -3\omega_{p3}$$

$\frac{A_{03}}{K} = 2$

$$s_2 = +\omega_{p3}\left[-1+\left(\frac{1}{2}+j\frac{\sqrt{3}}{2}\right)^2\right] = +\omega_{p3}\left(j\sqrt{3}\right)$$

$$s_3 = +\omega_{p3}\left[-1+\left(\frac{1}{2}-j\frac{\sqrt{3}}{2}\right)^2\right] = +\omega_{p3}\left(-j\sqrt{3}\right)$$

$$|L_u| = \frac{A_{03}|K}{|1+s(\omega_{p3})^3}$$

$$|L_u|_{\omega=\omega_{p3}\frac{\sqrt{3}}{2}} = \frac{8}{\left|1+j\frac{\sqrt{3}\cdot\omega_{p3}}{\omega_{p3}}\right|^3} = \frac{8}{|1+j\sqrt{3}|^3} = \frac{8}{8} = 1$$

$$\angle L_u = -180^\circ$$

$$\angle L_h \Big|_{\omega = \sqrt{3}\omega_{p3}} = -3\tan^{-1}\left(\frac{\sqrt{3}\omega_{p3}}{\omega_{p2}}\right) = -3 \times 60^\circ = -180^\circ$$